

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I S. Rohith (192521127) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Rohith

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Application Based questions

1. Auto scaling

- * Automatically increase the number of servers when traffic is high.
- * Removes extra servers when traffic decreases.
- * Ensures enough resources are available during seasonal sales.

Load balancing.

- * Distributes incoming user requests across multiple servers.
- * Prevents any single server from becoming overloaded.
- * Redirects traffic to healthy servers if one server fails.

Reducing downtime and improving user experience.

- * Prevents server crashes during heavy traffic.
- * Keeps the website available even during peak sales.

Role of monitoring tools.

- * Continuously monitor CPU, memory, networking traffic and server health.
- * Detect performance issue in real time.

Real world Ecommerce Example .

- * During Amazon prime day or flipkart Big billion days, million of users visit the website .
- * Auto scaling adds more servers to handle the increased traffic .

2. cloud storage for backup durability .

- * cloud storage is used to store large amounts of backup data safely .
- * Data is stored on multiple servers to prevent data loss .

Data redundancy .

- * Multiple copies of the same data are stored in different locations .
- * Protects data from hardware failures and natural disasters .

Encryption and access control .

- * Encryption converts data into a secure format and unauthorized users cannot read .
- * Data is encrypted during storage and transmission .

Cost Advantages .

- * No need to purchase expensive storage hardware .
- * Reduces maintenance and infrastructure costs .

Comparison with traditional storage.

- * traditional storage requires physical servers and regular maintenance.
- * cloud storage provide automatic backups and easy scalability.

3. Content delivery networks (CDNs)

- * CDNs store copies of app content on multiple servers worldwide.
- * users receive data from the nearest server.

Edge servers.

- * Edge servers are located close to users in different regions.
- * They deliver content faster by reducing the travel distance of data.

Global availability

- * cloud providers have data centers in multiple regions.
- * applications can run from different locations worldwide.

Benefits.

- * Faster loading of website and mobile apps
- * Reduced network congestion and lower latency.

Real time Example

- * Netflix uses CDNs to stream movies and TV shows smoothly worldwide.
- * PUBG mobile and other online games use edge servers to reduce lag and improve gameplay.

4. Regulatory compliance and auditing.

- * cloud providers follow industry security standards and regulations.
- * They provide audit reports and compliance certification.

Role of logging, monitoring and encryption

- * Logging records all user activities and system events.
- * Monitoring detects suspicious activities and security threats in real time.

Data sovereignty

- * Data is stored in specific geographic regions as required by law.
- * Cloud providers allow organizations to choose the data center location.

Benefits

- * Protects confidential customer information.
- * Facilitates secure financial transactions.

Example .

- * bank stores customer account details and online transaction records in a secure cloud data center located within its country.

5. Containers vs virtual machines

- * containers share the host operating system while virtual machines have their own operating system.
- * Containers start faster and use fewer resources.

Container Orchestration .

- * Kubernetes automates the deployment of containers.
- * It automatically scales applications based on demand.

Portability across cloud environment

- * Containers package the application with all its dependencies.
- * The same container can run on different cloud platforms without changes.

Benefits .

- * Faster application deployment .
- * Better resource utilization .

microservices Example .

- * Each service runs in its own container .
- * This ensures the application remains fast, reliable and highly available .